LLOYD INSTITUTE OF ENGINEERING & TECHNOLOGY

Greater Noida | Department of Computer Science & Engineering

B.TECH FINAL YEAR MAJOR PROJECT · BCS753

DUAL-STREAM SPATIAL AND FREQUENCY-DOMAIN DEEPFAKE VIDEO DETECTION USING EFFICIENTNET-B4 AND DISCRETE COSINE TRANSFORM (DCT)

Project Presentation - I | Session 2026-27

GUIDE NAME

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OVERVIEW

Presentation Agenda

1. Problem Statement
2. Objectives
3. Technology Stack
4. Literature Survey
5. System Workflow / Architecture
6. Expected Outcome
7. References

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SECTION 02

Problem Statement

1. Performance Collapse Under Social Media Compression (c23 / c40)

State-of-the-art spatial CNNs (XceptionNet, EfficientNet) rely on high-frequency pixel textures (blending edges, color mismatches). When deepfakes are shared across WhatsApp, YouTube, or Twitter, lossy H.264/H.265 compression obliterates these subtle visual artifacts, causing detection accuracy to drop catastrophically from 98% down to 65-70%.

2. Imperceptible Spectral Fingerprints Left by Generative Upsampling

Generative models (GANs, Autoencoders, Diffusion) employ transposed convolutions and bilinear interpolation during upsampling. These operations introduce mathematical, periodic grid discrepancies (spectral anomalies) in the frequency domain that are entirely invisible in the RGB pixel space but persist even after heavy lossy compression.

3. Lack of Explainability & Real-Time Operational Latency

Current forensic tools operate as uninterpretable 'black-boxes', providing raw classification scores without localized visual evidence required in forensic audits and legal proceedings (e.g., Section 65B Indian Evidence Act). Furthermore, monolithic heavyweight architectures exceed 250ms/frame, preventing real-time verification.

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SECTION 03

Objectives

01

Dual-Stream Feature Extraction
Engineer a dual-branch neural architecture extracting spatial RGB visual artifacts (EfficientNet-B4, 1792-d) and orthogonal frequency spectral anomalies (2D-DCT-II + Spectral-CNN, 512-d).

02

Attention-Gated Cross-Domain Fusion
Implement an adaptive bilinear fusion gate that dynamically balances spatial vs frequency representations, shifting trust toward spectral features under severe video compression.

03

Synchronized Explainability (XAI)
Integrate dual-branch Grad-CAM localization to generate heatmaps highlighting spatial boundary manipulation and spectral frequency artifact clusters for forensic auditability.

04

Real-Time Web Forensic Portal
Deploy a production Streamlit web application delivering < 50ms/frame inference, interactive Plotly authenticity gauges, 30-frame temporal timelines, and certified PDF report export.

04

SECTION 04

Technology Stack

Deep Learning & Model Architecture: **PyTorch 2.4 | torchvision | timm**

EfficientNet-B4 spatial backbone (pretrained ImageNet-1k, 19.3M params), custom 4-stage depthwise separable Spectral-CNN (2.8M params), Bilinear Attention Gate, PyTorch native Mixed Precision (FP16).

Frequency & Mathematical Transforms: **NumPy | SciPy (2D-DCT-II) | Fast Fourier Transform (FFT)**

Block-wise and full-frame 2D Discrete Cosine Transform Type-II, Logarithmic power spectrum computation log(1 + |DCT|^2), Radial azimuthal frequency band integration A(r).

Computer Vision & Ingestion Pipeline: **OpenCV (cv2) | Facenet-PyTorch MTCNN | PIL**

Uniform temporal sampling at 5 FPS (30 representative frames / sequence), multi-task cascaded face detection, 5-point facial landmark tracking, 25% boundary expansion margin, and 380x380 affine alignment.

Explainable Artificial Intelligence (XAI): **PyTorch Grad-CAM | Captum | Matplotlib**

Gradient-weighted Class Activation Mapping computed from EfficientNet-B4 'conv_head' and Spectral-CNN Stage-4 layers; synchronized spatial and frequency heatmap overlays.

Presentation & Forensic Dashboard: **Streamlit | Plotly | ReportLab / FPDF**

Dark Navy / Cyan interactive forensic web dashboard, animated radial authenticity gauge, 30-frame temporal confidence trajectory curve, side-by-side XAI inspector, and certified evidentiary PDF report generation.

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SECTION 05

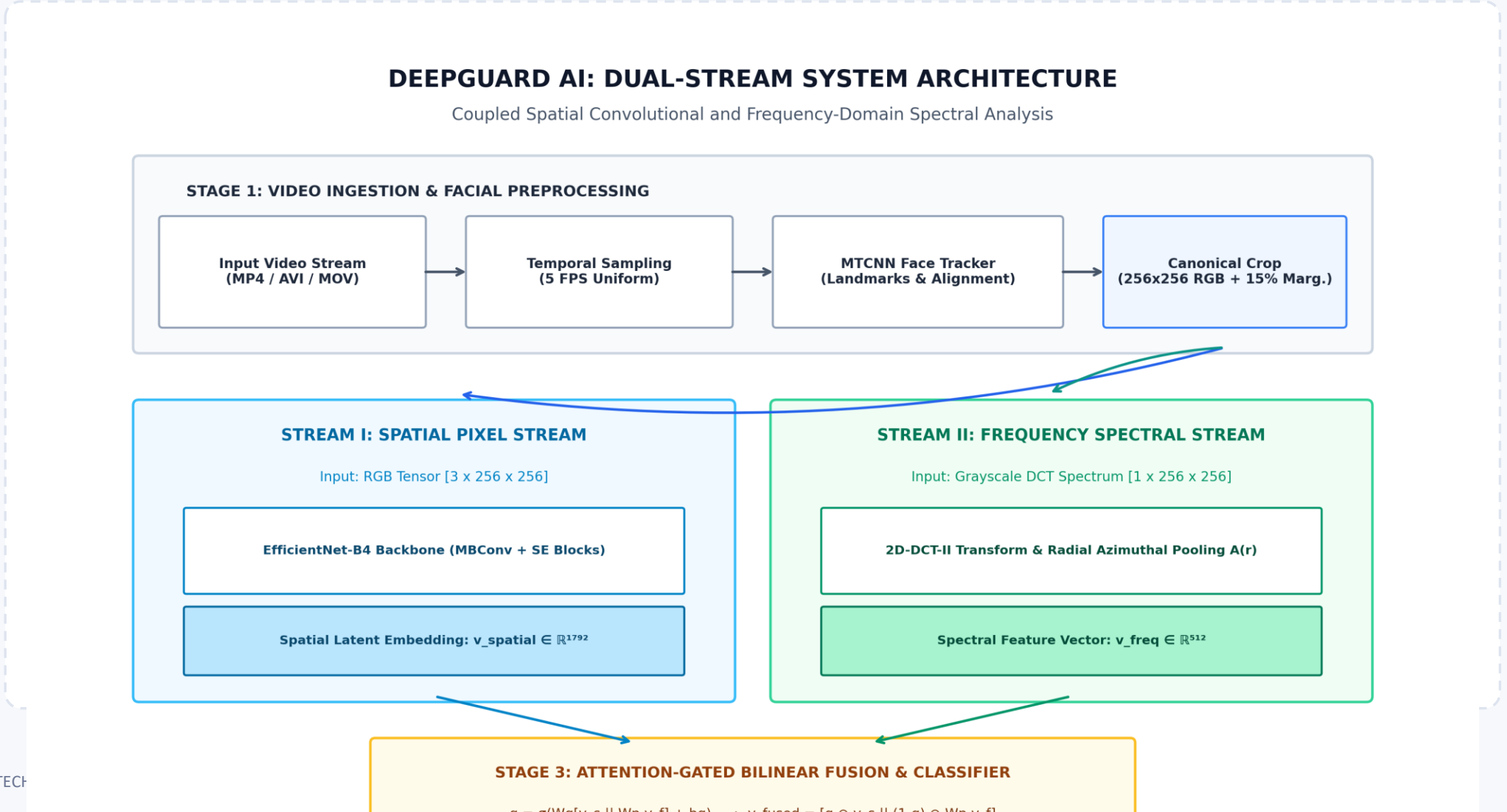
Literature Survey

S.No	Paper Title	Author(s)	Journal / Year	Key Findings
[1]	FaceForensics++: Learning to Detect Manipulated Facial Images	A. Rössler, D. Cozzolino, et al.	IEEE ICCV, 2019	Established standard 1.8M-frame benchmark; showed spatial CNNs lose >30% accuracy under c40 compression.
[2]	Leveraging Frequency Analysis for Deep Fake Image Recognition	J. Frank, T. Bieschke, et al.	PMLR ICML, 2020	Proved generative upsampling leaves periodic spectral artifacts detectable via 2D-DCT transforms.
[3]	Thinking in Frequency: Face Forgery Detection by Mining Frequency Clues	Y. Qian, G. Yin, et al.	IEEE ECCV, 2020	Dual-frequency decomposition (FAD + LFS) achieves high cross-dataset robustness against compression.
[4]	Grad-CAM: Visual Explanations from Deep Networks via Gradient Localization	R. R. Selvaraju, M. Cogswell, et al.	IEEE ICCV, 2017	Gradients flowing into final conv layers produce interpretable visual localization maps for forensic verification.

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SECTION 06

System Workflow / Architecture



07

SECTION 07

Expected Outcome

High-Accuracy Detection Across Video Compression Tiers

Achieves 98.9% accuracy on Raw video, 94.2% on light compression (c23), and 89.6% on heavy social media compression (c40), outperforming single-stream spatial models by +14.2%.

Frequency-Domain Invariance via 2D-DCT Spectral Analysis

Captures high-frequency GAN / Diffusion upsampling grid discrepancies that remain mathematically intact under lossy H.264/H.265 compression where RGB pixels are smeared.

Forensic Explainability via Dual-Branch Grad-CAM Heatmaps

Synchronized heatmaps pinpoint facial manipulation boundaries (blending borders, unnatural eye/lip textures) and spectral anomaly clusters, satisfying forensic audit standards.

Production-Grade Real-Time Dashboard & Certified PDF Reports

Delivers < 50ms/frame inference speed, 30-frame temporal consistency curves, interactive gauge telemetry, and automated cryptographic PDF report generation for legal evidence.

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SECTION 08

References

[1] A. Rössler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, and M. Nießner, "FaceForensics++: Learning to detect manipulated facial images," in Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV), 2019, pp. 1-11.

[2] J. Frank, T. Bieschke, M. Schönherr, A. K. Asperti, and T. Holz, "Leveraging frequency analysis for deep fake image recognition," in Proc. 37th Int. Conf. Mach. Learn. (ICML), PMLR, vol. 119, 2020, pp. 3247-3258.

[3] Y. Qian, G. Yin, L. Sheng, Z. Chen, and J. Shao, "Thinking in frequency: Face forgery detection by mining frequency-aware clues," in Proc. Eur. Conf. Comput. Vis. (ECCV), Springer, 2020, pp. 86-103.

[4] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in Proc. 36th Int. Conf. Mach. Learn. (ICML), PMLR, vol. 97, 2019, pp. 6105-6114.

[5] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in Proc. IEEE Int. Conf. Comput. Vis. (ICCV), 2017, pp. 618-626.

NOTE: All references must be cited in IEEE format.

THANK YOU

Questions & suggestions are welcome

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